

DIY Rifle Level - BOM and Wiring Guide

XIAO nRF52840 Sense, 3 LED level indicator, calibration button, LiPo power

Build assumptions

- Rifle-mounted electronic level using the Seeed Studio XIAO nRF52840 Sense.
- The Sense board has the onboard 6-axis IMU used for roll/cant sensing.
- 3 external LEDs: left red, center green, right blue.
- One 6x6x7 mm momentary tactile calibration button.
- One 1S 3.7V 300 mAh 502035 LiPo battery and one SS12D00G4 3-pin SPDT slide switch used as a simple true-off switch.
- Simple repairable wiring: one common ground bus for LED negatives and button ground.

Pin assignment

D0	Left red LED
D1	Center green LED
D2	Right blue LED
D3	Calibration button
GND	Common ground bus
BAT+	Switched LiPo positive
BAT- / GND	LiPo negative

Core wiring summary

D0 -> 330 ohm -> red LED +
D1 -> 330 ohm -> green LED +
D2 -> 330 ohm -> blue LED +
LED - legs -> ground bus -> GND
D3 -> button -> ground bus
LiPo + -> SS12D00G4 center -> BAT+
LiPo - -> BAT- / GND

Important

- Do not connect the LiPo to 5V or 3V3. Use BAT+ and BAT- / GND.
- Do not connect 3V3 to GND through the switch.
- Each LED needs its own resistor. Do not skip the resistors.
- With this simple true-off switch, keep the switch ON while charging through the board.

Verify pad names against the board silkscreen before soldering. The back-side image in this guide is only a pad reference; the wiring diagram shows the final connections.

Bill of Materials (BOM)

Recommended practical parts for the rifle level build

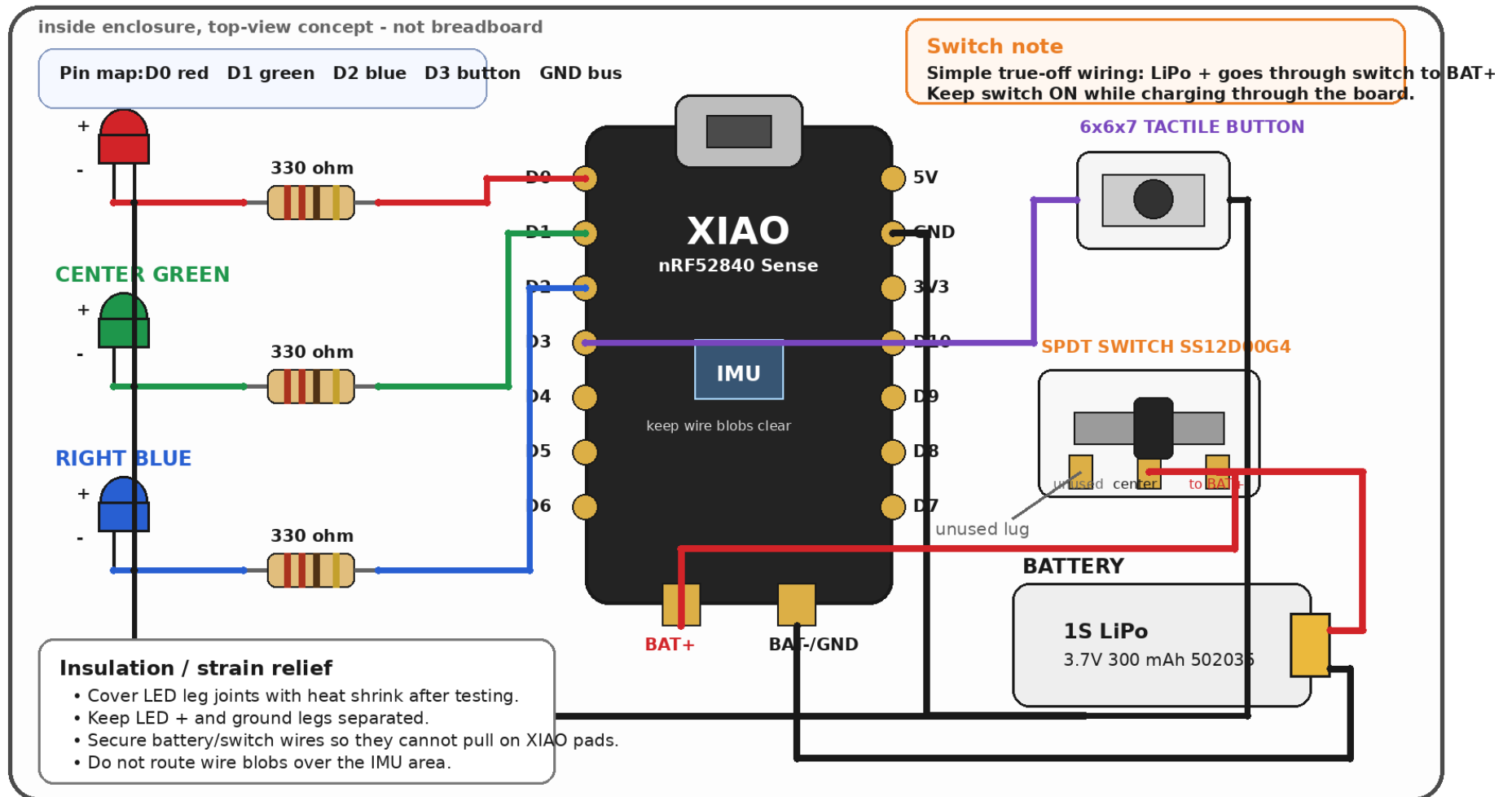
Item	Qty	Recommended spec	Purpose / notes
Seeed Studio XIAO nRF52840 Sense	1	Sense version	Main controller, Bluetooth, and onboard IMU.
1S LiPo battery	1	3.7V, 300 mAh, 502035 size, prewired 28 AWG leads	Main power. Purchased from Amazon listing B0CNLPWPBQ. Existing 28 AWG leads are fine for this low-current build.
SPDT slide switch	1	SS12D00G4, 3-pin mini slide switch	Center pin gets LiPo +. One outer pin goes to BAT+. Other outer pin unused.
Tactile calibration button	1	6x6x7 mm momentary tactile switch	D3 to button, other side to ground bus. Code uses INPUT_PULLUP.
Red LED	1	3 mm LED	Left/cant-left indicator. Positive leg goes through resistor to D0.
Green LED	1	3 mm LED	Center/level indicator. Positive leg goes through resistor to D1.
Blue LED	1	3 mm LED	Right/cant-right indicator. Positive leg goes through resistor to D2.
LED resistors	3	330 ohm, 1/8W or 1/4W	One resistor per LED. 220 ohm can be tested later if the LEDs are too dim.
Wire	as needed	28 AWG stranded silicone, tinned copper preferred	LED wires, button wires, switch wire, and common ground bus.
Heat shrink tubing	as needed	Small diameter tubing sized for LED legs and resistor joints	Insulate LED leg joints and exposed resistor joints after testing.
Neutral-cure silicone or flexible adhesive	small amount	Electronics-safe / neutral-cure	Optional strain relief after testing. Do not cover the IMU area with blobs.
M2 heat-set inserts	4	M2 x 4 x 3.2 mm	For enclosure and lid mounting bosses.
M2 countersunk screws	4	M2 x 4 mm	For enclosure and lid.
M3 heat-set inserts	2	M3 x 6 x 5 mm	For clamp/mount hardware.
M3 countersunk screws	2	M3 x 10 mm	For clamp.
Transparent filament light tube	1 small piece	Clear/transparent 1.75 mm filament, about 5 mm long	Short light tube for the onboard charge/status LED window. Trim to fit the lid.
3D printed enclosure, lid, and clamp	1 set	Your final PETG-CF / PA6-CF / chosen material design	Compact side/cantilever mount enclosure.

Illustrated Final Wiring Diagram

Use this as the enclosure wiring plan - not a breadboard layout

Final Wiring Diagram - XIAO nRF52840 Sense Rifle Level

Common ground bus + SS12D00G4 SPDT slide switch + Left red / Center green / Right blue



Wiring Steps

Recommended soldering order for a clean, repairable enclosure build

1. Pre-test the board

Confirm the XIAO programs and runs before soldering the LiPo.

2. Prepare the LEDs

Bend the LED legs 90 degrees to match the enclosure. Trim each leg so only about 4-6 mm remains past the bend. Identify the negative leg: short leg / flat side.

3. Add one resistor per LED

Solder a 330 ohm resistor to each LED positive leg or place the resistor inline between the XIAO pin and the LED positive leg.

4. Make the common ground bus

Run one black 28 AWG wire along the LED row. Solder the red, green, and blue LED negative legs to this bus. Also connect one side of the calibration button to this same ground bus.

5. Wire the LED signals

D0 goes to the red LED resistor. D1 goes to the green LED resistor. D2 goes to the blue LED resistor.

6. Wire the calibration button

D3 goes to the other side of the 6x6x7 mm tactile button. The button ground side goes to the common ground bus. The code should use INPUT_PULLUP.

7. Wire the battery switch

LiPo red wire goes to the center pin of the SPDT switch. One outer switch pin goes to BAT+. Leave the other outer pin unused. If the ON direction feels backward, move the BAT+ wire to the other outer pin.

8. Wire battery negative

LiPo black wire goes to BAT- or GND on the XIAO. BAT- and GND share the same ground reference.

9. Insulate and support

After testing, cover LED leg joints and exposed resistor joints with heat shrink. Add strain relief so battery and switch wires do not pull on XIAO pads.

10. Install the light tube

Cut a clear/transparent filament piece to about 5 mm and test-fit it in the lid light-tube hole.

Common ground bus concept

XIAO GND -> one black ground bus wire -> LED negatives + button ground



Final Checks Before Power

Use a multimeter before plugging in the battery

Multimeter checklist

- Continuity check: BAT- and GND should beep / show near 0 ohms.
- Short check: BAT+ to GND should NOT beep.
- Switch OFF: LiPo + should NOT connect to BAT+.
- Switch ON: LiPo + should connect to BAT+ through the SPDT switch.
- Each LED positive path has a resistor between the XIAO pin and LED positive leg.
- LED negatives and button ground all connect to the common ground bus.
- D3 connects only to the button signal side, not directly to GND unless the button is pressed.
- No exposed LED leg, resistor leg, switch lug, or battery wire can touch another pad or wire.
- Battery wires have strain relief so they cannot rip pads off the XIAO.
- The IMU area on the XIAO is not covered by heavy glue blobs, wire bundles, or tape stacks.
- The transparent filament light tube is about 5 mm long and aligned with the onboard charge/status LED.

Board pad reference

Use the board silkscreen / Seeed pad image to verify pad names before soldering.

